

LABID AL NAHIYAN

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EDUCATION

- **Bangladesh University of Engineering and Technology** Feb 2020 - March 2025
BSc in Computer Science and Engineering CGPA : 3.62
- **Notre Dame College** July 2017 - April 2019
Higher Secondary Certificate (General scholarship recipient) GPA : 5.00

THESIS

Digging into a New Contextually Appropriate Loss Function for Dense Crowd Tracking

Supervisor: Prof. A. B. M. Alim Al Islam, BUET

- Proposed a novel size-aware, gradient-stabilized loss function tailored for tracking small, overlapping objects in dense crowds.
- Integrated the loss into YOLO-based models and achieved higher mAP@50 and F1-score on JHU-CROWD++ and SCUT-HEAD datasets compared to traditional CIoU, DIoU, and GIoU losses.

RESEARCH INTERESTS

- Machine Learning, Computer Vision, Natural Language Processing

PROJECTS

NerdHerd - All in one Educational Platform [SvelteKit, TailwindCSS, PostgreSQL] [Github](#)

- Built an all-in-one student platform with video classes, quizzes, and forums. Handled frontend and full-stack integration.

Shop Genie - AI-Driven E-commerce Platform [SvelteKit, LLM, RAG] [Github](#)

- Built an AI shopping assistant using LLMs for natural language queries and product retrieval across 10k+ items.

Bidirectional Autonomous Robot with Obstacle Avoidance [Arduino] [Website](#)

- Used Hierarchical Q-learning for intelligent movement. Built custom chassis and integrated policy adaptation.

Subset of C Compiler [C, Bison, YACC, Flex] [Github](#)

- Implemented lexical analysis, parsing, symbol table, and code generation.

Football Player Management [Java, JavaFX] [Github](#)

- Simulated transfer market with network-based player trading.

TECHNICAL SKILLS

Programming Languages	C++, Java, SQL, Python, HTML, CSS, TypeScript, Go
Frameworks	React.js, Next.js, Svelte, Tailwind CSS, Express.js, Postgres
Tools	GitHub, Postman, VS Code, Docker, Linux, Windows

ACHIEVEMENTS

- Champion, OpenAPI Hackathon, 11th National ICT Fest 2024 (IUT)
- 7th Place, SEC Inter University Junior Programming Contest 2022 (Team: BUET Fast Furious)